Politecnico di Milano

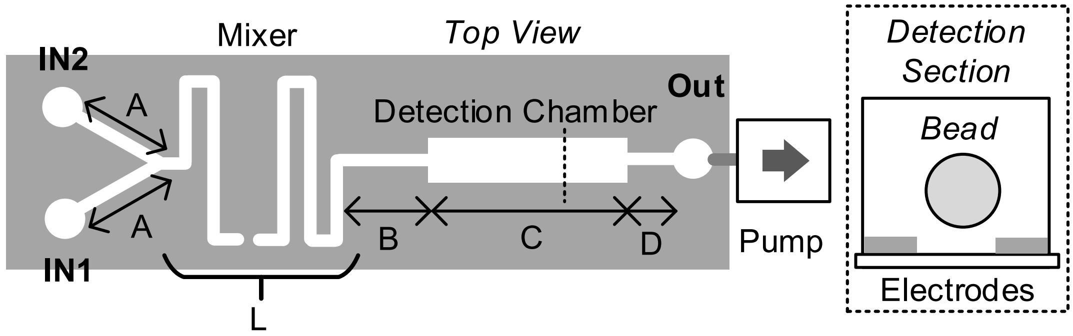
Biochip A.Y. 2020/2021 - M. Carminati

Exam of 20.01.2022

Test duration: 2 hours – No support material (textbooks, laptops, smartphones, notes) admitted.
All the answers must be motivated. Beyond the final numerical results, all the expressions must be provided.

Exercise 1

Consider the following microfluidic device for impedimetric counting of individual antibodies in a liquid sample. The input sample (IN1) is mixed with functionalized beads (IN2) in a passive mixer. The flow is driven in aspiration by a peristaltic pump. The channel height h is fixed.



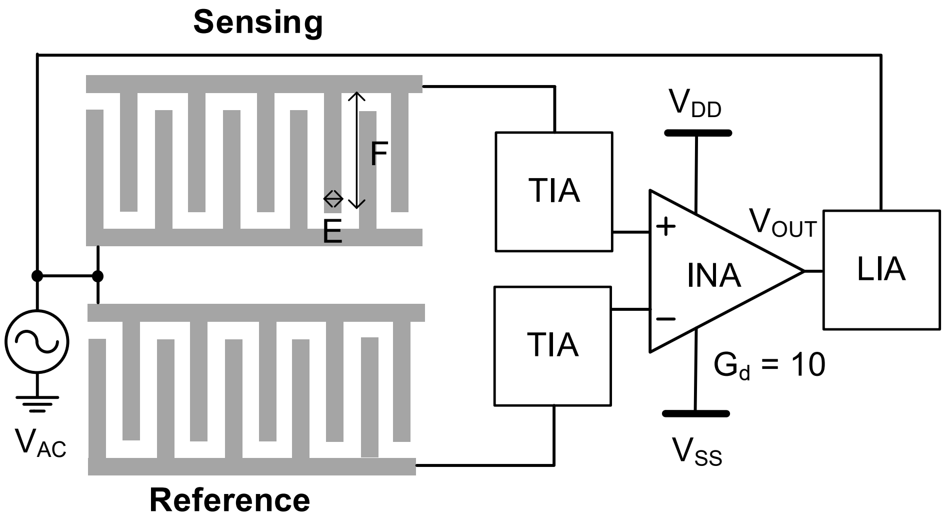
Data:

h = 10 μm
$w_A = w_B = w_D = 10$ μm
$w_C = 30$ μm
A = 1 mm
B = 1 mm
C = 2 mm
D = 0.5 mm
1atm = 101 kPa
$\eta_{water} = 10^{-3}$ Pa·s

a) Draw the electrical fluidic equivalent of the system. Choosing a channel width in the range from 10 to 20 μm, size the maxim length L of the mixer section compatible with a max. resistance of this section of 10^{16} Pa·s/m^3.
b) Assuming P_{IN1} = 1 atm, P_{IN2} = 1 atm, P_{OUT} = 0.5 atm find the fluid speed in all sections of the device.
c) If the peristaltic pump has 3 rollers and is rotating at a speed of 200 rpm, find a solution to attenuate by a factor 10 the ripple in the generated flow.
d) Discuss the impact of an increase on temperature on the passive mixer and propose an alternative mixing scheme using a piezo actuator.
e) Compare additive manufacturing and replica molding for the fabrication of chips as the one here considered.

Exercise 2

Now focus on the detection section of the chip. Antigens are captured on the sensing group of interdigitated planar electrodes (5 fingers of width E and length F and gap 2 μm) and their effect is enhanced by capturing functionalized plastic beads of 1 μm diameter. Their presence is detected by means of a differential impedance scheme, with an INA with a differential gain $G_d = 10$, i.e. $V_{OUT} = 10\cdot(V_+ - V_-)$



Data:

E = 2 μm
F = 50 μm
5 exposed fingers

V_{AC} = 10 mV (peak)
V_{DD} = +5 V
V_{SS} = -5 V

$\varepsilon_0 = 8.86\cdot10^{-12}$ F/m
$\varepsilon_{r\_water} = 80$
δ_{PBS} = 1.5 S/m
C_{DL} = 0.1 pF/μm^2

a) Find the equivalent impedance between the electrodes, in the absence of antibodies and considering only the area of the 5 fingers, assuming PBS as buffer liquid and a value of the solution resistance R_{SOL} = 20 kΩ. Find also the optimal sensing frequency.
b) Draw a detailed scheme of the transimpedance amplifier (TIA) and size the value of its feedback resistor R_F.
c) Draw a detailed scheme of the lock-in block (LIA) and size the bandwidth of the LPF in order to have an input resolution of 3 pA$_{RMS}$ (considering only the noise of resistors).
d) How could the system be adapted to detect a different molecule?
e) Propose an alternative differential sensing scheme that does not requires the INA and compare the two options.